

Simple Analytical Magnetic Mirror Orbit Benchmark

1. Motivation and purpose

A magnetic mirror confines charged particles by using a magnetic field that is weaker in the central region and stronger near the ends. A charged particle moving in such a field executes rapid gyromotion around the local magnetic-field direction while also moving more slowly along the field. If the magnetic field changes gradually over one Larmor orbit, the particle approximately conserves its first adiabatic invariant,

$$\mu = \frac{mv_{\perp}^2}{2B},$$

where m is the particle mass, $v_{\perp}$ is the velocity perpendicular to the local magnetic field, and $B = |\mathbf{B}|$ is the magnetic-field magnitude. Conservation of μ causes particles with sufficiently large pitch angle to reflect from the high-field end regions. Particles with too small a pitch angle pass through the end mirror and are lost.

The purpose of this benchmark is to test a particle orbit pusher in a deliberately simple, transparent magnetic-mirror geometry. It uses an analytical axisymmetric magnetic field composed of a uniform guide field plus two identical end solenoids. The parameters were intentionally chosen so that the ion Larmor radius is large enough to be clearly visible in the trajectory plots. This makes the calculation useful as a visual and pedagogical benchmark, although it also makes the particle motion less adiabatic than in a realistic high-field mirror.

2. Magnetic-field model

The magnetic field is represented as a uniform guide field plus two end solenoids centered at $z = \pm Z_{\text{sol}}$. In the calculation,

$$\begin{aligned} B_{\text{guide}} &= 0.08 \text{ T} \\ B_{\text{mirror}} &\approx 0.40 \text{ T} \end{aligned} ,$$

with end solenoids centered at

$$z = \pm 2.0 \text{ m} .$$

The effective mirror ratio is therefore approximately

$$R_m = \frac{B_{\text{max}}}{B_{\text{launch}}} \approx 4.57.$$

The field is axisymmetric, but the particle pusher uses Cartesian components

$$B_x, B_y, B_z.$$

The code first constructs an analytical on-axis field $B_0(z)$. The off-axis field is then approximated using the lowest-order divergence-free expansion for an axisymmetric field:

$$B_x = B_r \frac{x}{r}, B_y = B_r \frac{y}{r},$$

with

$$B_r(r, z) \simeq -\frac{r}{2} \frac{dB_0}{dz},$$

and

$$B_z(r, z) \simeq B_0(z) - \frac{r^2}{4} \frac{d^2 B_0}{dz^2}.$$

Thus, although the field is generated from an axisymmetric construction, the orbit pusher sees the fully three-dimensional Cartesian field vector

$$\mathbf{B}(x, y, z) = B_x \hat{x} + B_y \hat{y} + B_z \hat{z}.$$

This keeps the benchmark compatible with a 3-D particle pusher while retaining a simple analytical field model.

3. Particle model and numerical method

The test particles are 14 keV deuterons. Their speed is set by

$$v = \sqrt{\frac{2E}{m_D}},$$

where $E = 14$ keV and m_D is the deuteron mass. Particles are launched near the midplane at $x_0 = 0.10$ m, $y_0 = 0, z_0 = 0$.

Several pitch angles are tested (in degrees):

15, 25, 30, 45, 60, 80

The orbit integration uses the Boris pusher. Since the electric field is zero in this benchmark, the Boris update reduces to a magnetic rotation of the particle velocity followed by a position update. This is a standard and robust method for charged-particle orbit integration because it preserves particle speed well in static magnetic fields.

The guide field was deliberately chosen to be weak enough that the Larmor radius is visible. For the 60° particle, the launch Larmor radius is approximately

$$\rho_L \approx 0.24 \text{ m}.$$

This is large compared with the transverse scale of the plots, so the gyro-orbit is easy to see.

4. Mirror theory

For an adiabatic particle, the magnetic moment

$$\mu = \frac{mv_{\perp}^2}{2B}$$

is approximately conserved. Since the total kinetic energy is also conserved in a static magnetic field, an ion moving into a stronger magnetic field converts parallel kinetic energy into perpendicular kinetic energy. Reflection occurs when the parallel velocity goes to zero.

At launch,

$$v_{\perp 0} = v \sin \alpha_0,$$

where α_0 is the launch pitch angle. Conservation of μ gives the mirror condition

$$\frac{v_{\perp 0}^2}{B_0} = \frac{v^2}{B_{\text{mirror}}},$$

so that the particle mirrors when

$$B_{\text{mirror}} = \frac{B_0}{\sin^2 \alpha_0}.$$

If the required mirror field exceeds the maximum available field, the particle lies inside the loss cone and is lost. The loss-cone angle is

$$\sin^2 \alpha_{\text{LC}} = \frac{B_0}{B_{\text{max}}} = \frac{1}{R_m}.$$

For this benchmark,

$$R_m \approx 4.57,$$

so

$$\alpha_{\text{LC}} \approx \sin^{-1} \left(\frac{1}{\sqrt{4.57}} \right) \approx 27.9^\circ.$$

Therefore, mirror theory predicts that particles launched with pitch angles below about 28° should be lost, while particles launched above this angle should be trapped.

5. Numerical results

The representative particle calculations gave:

Pitch angle	Outcome
15°	lost
25°	lost
30°	trapped
45°	trapped
60°	trapped
80°	trapped

This agrees with the theoretical loss-cone estimate. The predicted loss-cone angle is approximately 27.9° , so the 15° and 25° particles are expected to escape, while the 30° , 45° , 60° , and 80° particles are expected to reflect.

The numerical and theoretical turning points also agree reasonably well for the trapped particles:

Pitch angle	$z_{\text{turn,theory}}$ [m]	$z_{\text{turn,num}}$ [m]
30°	1.816	1.833
45°	1.445	1.442
60°	1.126	1.120
80°	0.476	0.476

The agreement is best for the larger pitch angles. The 80° particle mirrors close to the center, where the magnetic field varies slowly, and the adiabatic theory works very well. The 30° particle is close to the loss cone and mirrors near the stronger-gradient end region, so it shows a somewhat larger discrepancy.

6. Magnetic moment conservation

The magnetic moment was monitored using

$$\frac{\Delta\mu}{\mu_0} = \frac{\mu(t) - \mu_0}{\mu_0}.$$

The maximum magnetic-moment deviations were not extremely small:

Pitch angle	max $ \Delta\mu/\mu_0 $
15°	1.57×10^{-1}
25°	8.93×10^{-2}
30°	1.89×10^{-1}
45°	1.75×10^{-1}
60°	1.50×10^{-1}
80°	3.50×10^{-2}

This is expected for this visualization-oriented setup. The Larmor radius was intentionally made large enough to be visible:

$$\rho_L \sim 0.24 \text{ m}$$

for the 60° ion. A large Larmor radius means the particle samples a significant variation of the magnetic field during one gyro-orbit. Therefore, the adiabatic condition

$$\rho_L \ll L_B$$

is only marginally satisfied. Here L_B is the characteristic magnetic-field gradient scale length. Because the orbit is not strongly adiabatic, μ is conserved only approximately.

This is an important feature of the benchmark. The same setup simultaneously tests both the qualitative magnetic-mirror effect and the expected breakdown of perfect magnetic-moment conservation when the gyro-radius is not very small.

7. Interpretation

The calculation behaves as expected. The loss cone predicted by adiabatic mirror theory occurs near

$$\alpha_{LC} \approx 28^\circ.$$

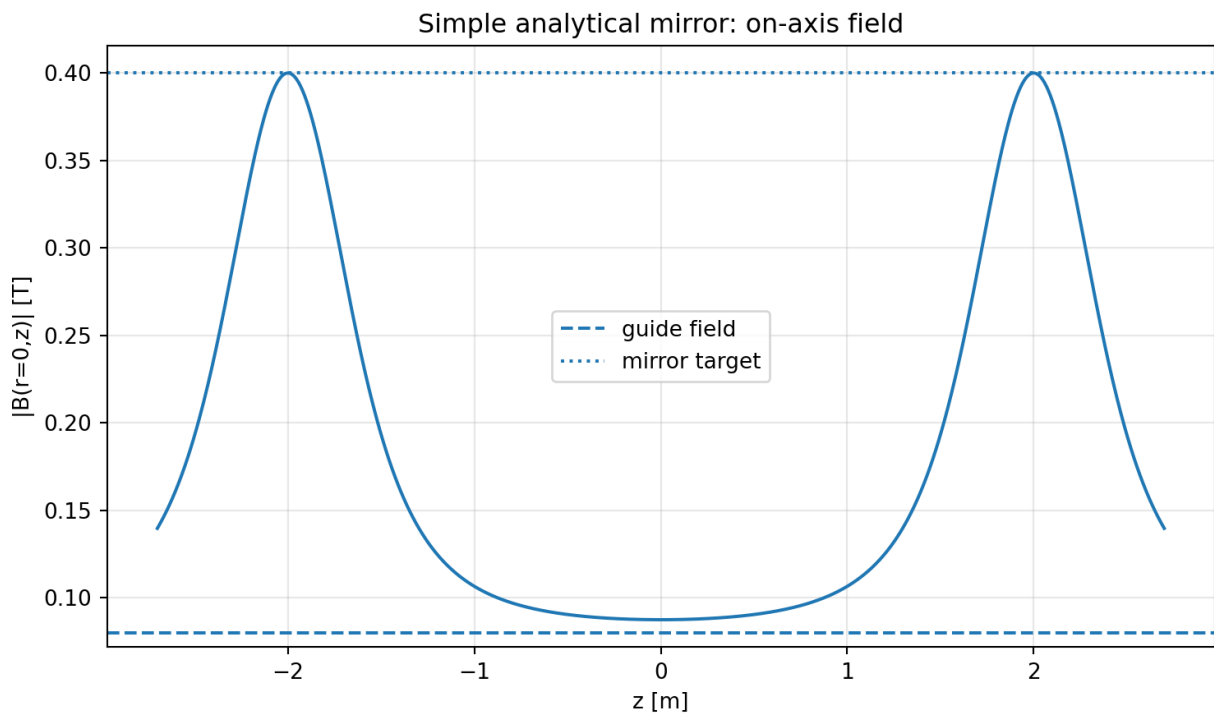
The 15° and 25° particles are lost, while the 30° and larger pitch-angle particles are trapped. The turning-point comparison shows that even with a deliberately large Larmor radius, the Boris pusher reproduces the basic mirror physics well.

The magnetic-moment diagnostic shows that the motion is not perfectly adiabatic. This is not a failure of the calculation; it follows from the design choice. The magnetic field was intentionally reduced so that the Larmor radius would be visible in the plots. A more realistic high-field mirror, or a lower-energy ion, would have a smaller Larmor radius and would show better conservation of μ , but the gyro-orbit would be much harder to see visually.

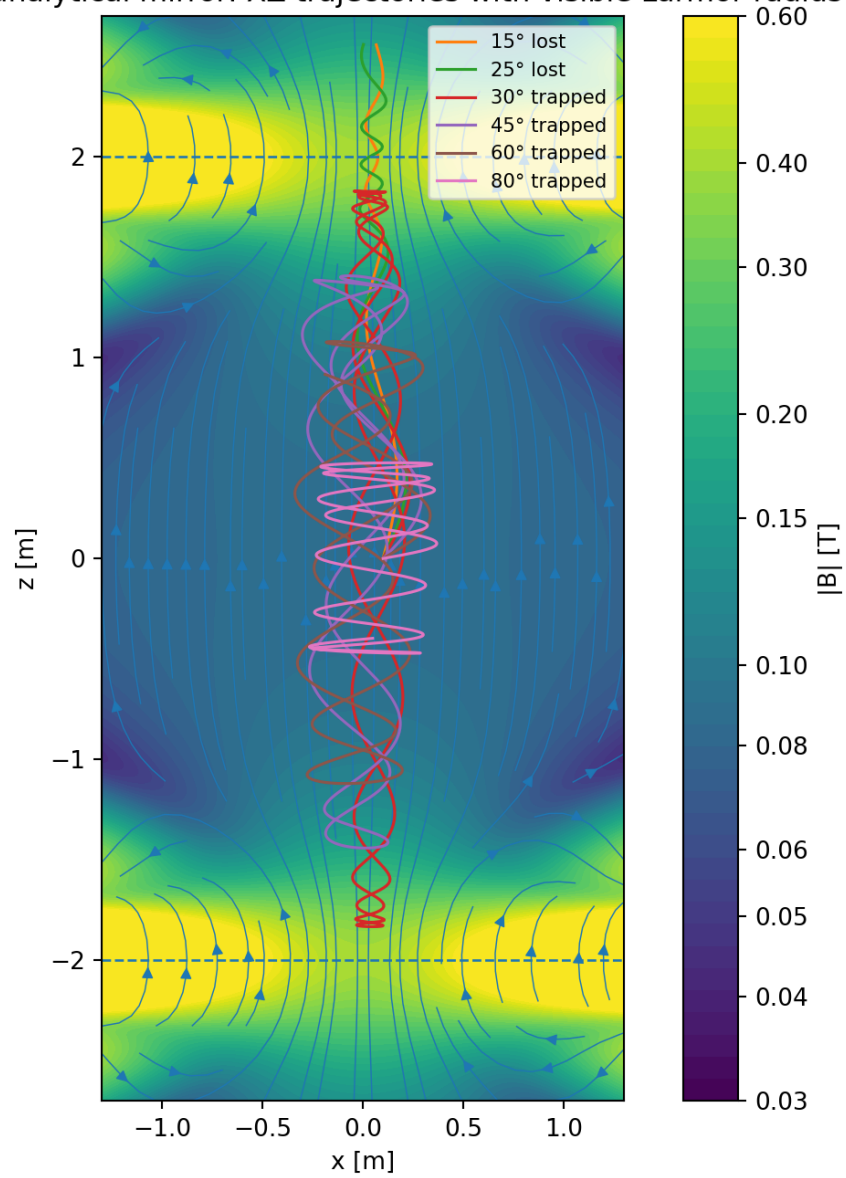
Thus, this simplified problem is useful as a first orbit-code benchmark because it shows, in one compact setup:

1. helical motion around the magnetic field,
2. loss-cone behavior,
3. mirror reflection,
4. agreement between numerical and theoretical turning points,
5. and the finite-Larmor-radius breakdown of exact magnetic-moment conservation.

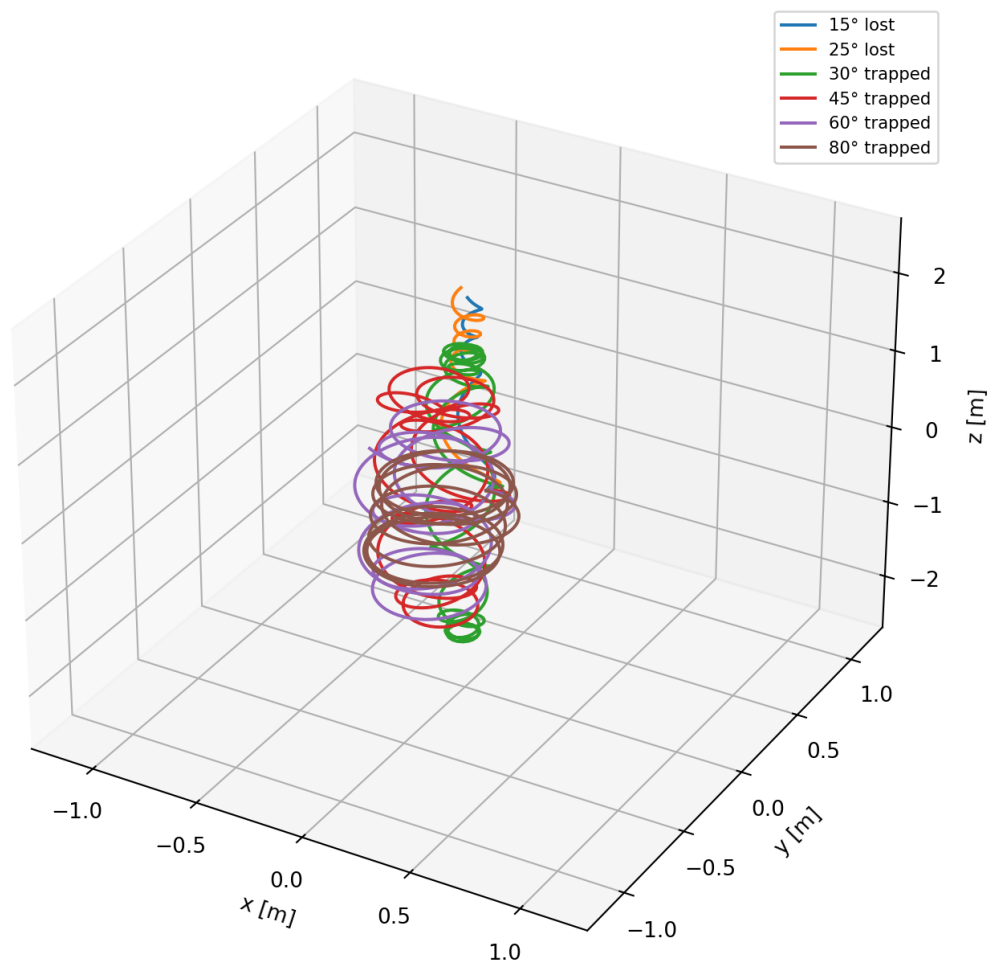
The model is not intended to reproduce a particular experimental machine. Its value is that the field is simple, analytical, and easy to control, while still exercising the main physics needed for later orbit calculations in more realistic magnetic geometries.

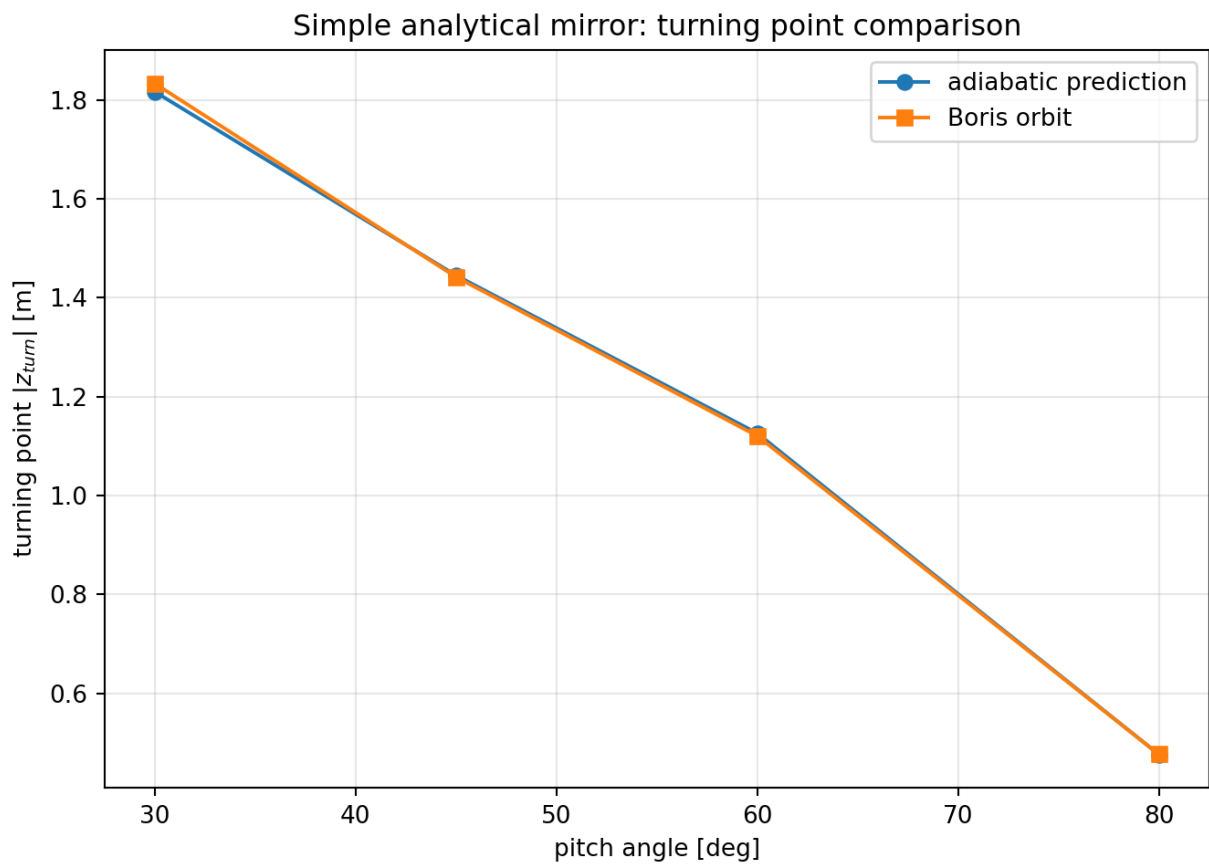


Simple analytical mirror: XZ trajectories with visible Larmor radius

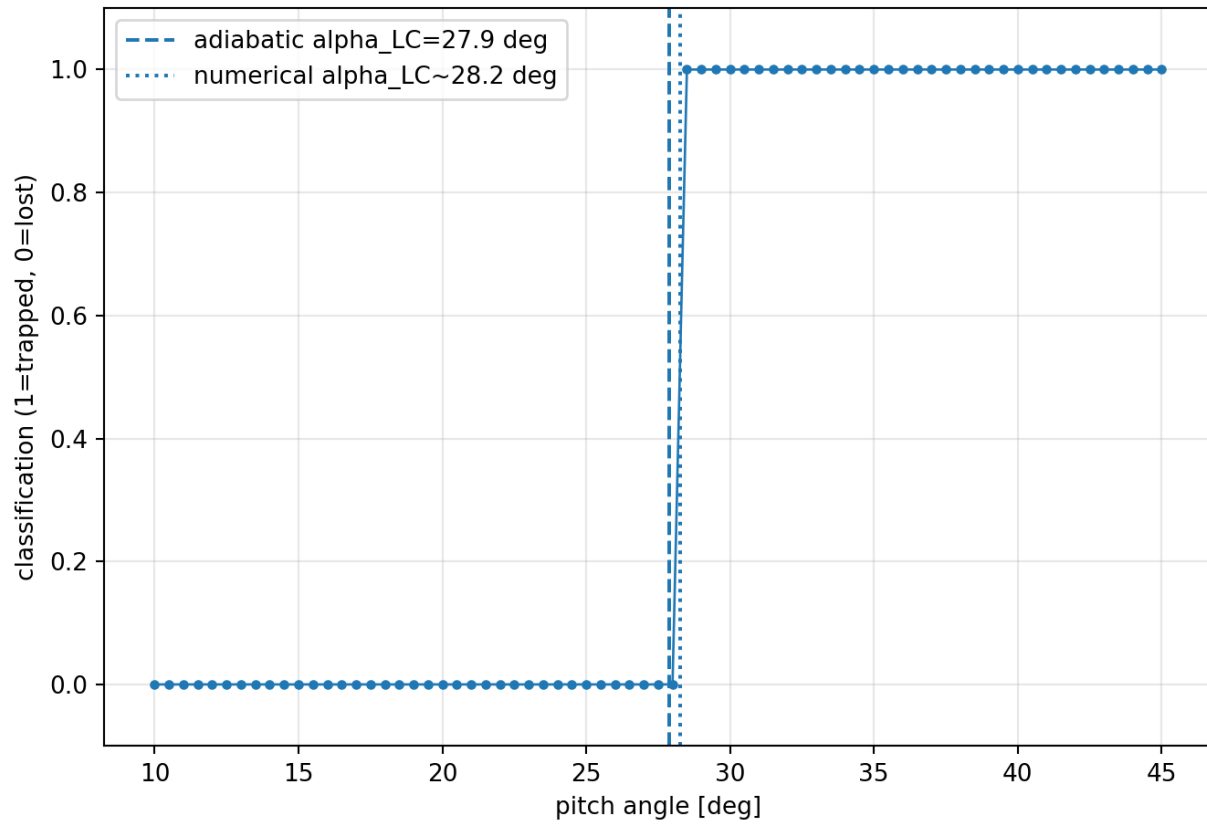


Simple analytical mirror: 3D orbits





Simple analytical mirror: loss-cone scan



Simple analytical mirror: magnetic moment conservation

